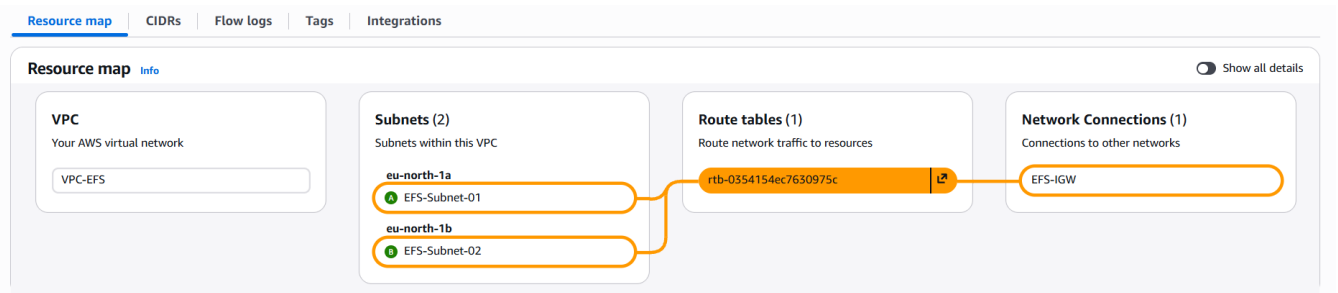


AWS Cloud Practical

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1. Elastic File System (EFS)

- ✓ Amazon Elastic File System (EFS) is a fully managed, scalable Network File System (NFS) for Linux-based workloads. Provides a shared file system that can be mounted on multiple EC2 instances at the same time.



- Launch an EC2 instance named Instance A inside the VPC-EFS.

The screenshot displays the AWS Management Console interface for an EC2 instance named 'Instance A' (ID: i-099a0f9a1178ff20b). The instance is in the 'Running' state and is located in the 'Europe (Stockholm)' region. The console shows various details about the instance, including its public and private IP addresses, DNS names, and associated VPC and subnet information.

Instance ID	Public IPv4 address	Private IPv4 addresses
i-099a0f9a1178ff20b	13.48.26.121 open address	10.0.0.10

IPv6 address	Instance state	Public DNS
-	Running	-

Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-10-0-0-10.eu-north-1.compute.internal	ip-10-0-0-10.eu-north-1.compute.internal	-

Answer private resource DNS name	Instance type	AWS Compute Optimizer finding
-	t3.micro	Opt-in to AWS Compute Optimizer for recommendations. Learn more

Auto-assigned IP address	VPC ID	Auto Scaling Group name
13.48.26.121 [Public IP]	vpc-0d32c10ca1a31558f (VPC-EFS)	-

IAM role	Subnet ID	Managed
-	subnet-090dbcfa222b1cb6b (EFS-Subnet-01)	false

IMDSv2	Instance ARN	
Required	arn:aws:ec2:eu-north-1:787169320097:instance/i-099a0f9a1178ff20b	

- Create an EFS file system inside the same VPC

Create file system ✕

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional
Name your file system.

My-EFS

Name can include letters, numbers, and +-=._:/ symbols, up to 256 characters.

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.

vpc-0d32c10ca1a31558f
VPC-EFS

Recommended settings

Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes
Transition into Infrequent Access (IA)	30 day(s) since last access	Yes
Transition into Archive	90 day(s) since last access	Yes
Transition into Standard	None	Yes
Automatic backups	Enabled	Yes
Encryption	Enabled	No

Cancel Customize Create file system

- While customizing the EFS configuration, AWS automatically creates mount targets in the selected subnets of the attached VPC.
- The security group attached to the EFS mount target must allow NFS (port 2049) with the source set to the EC2 instance's security group, because EFS communicates using the NFS protocol on port 2049.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Inbound rule 1

Delete

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source type [Info](#)

Source [Info](#)

Description - optional [Info](#)

EC2 instance Security Group

Elastic File System

File systems
Access points

AWS Backup [↗](#)
AWS DataSync [↗](#)
AWS Transfer [↗](#)

Documentation [↗](#)

Network

Virtual Private Cloud (VPC)

Choose the VPC where you want EC2 instances to connect to your file system.

vpc-0d32c10ca1a31558f
VPC-EFS

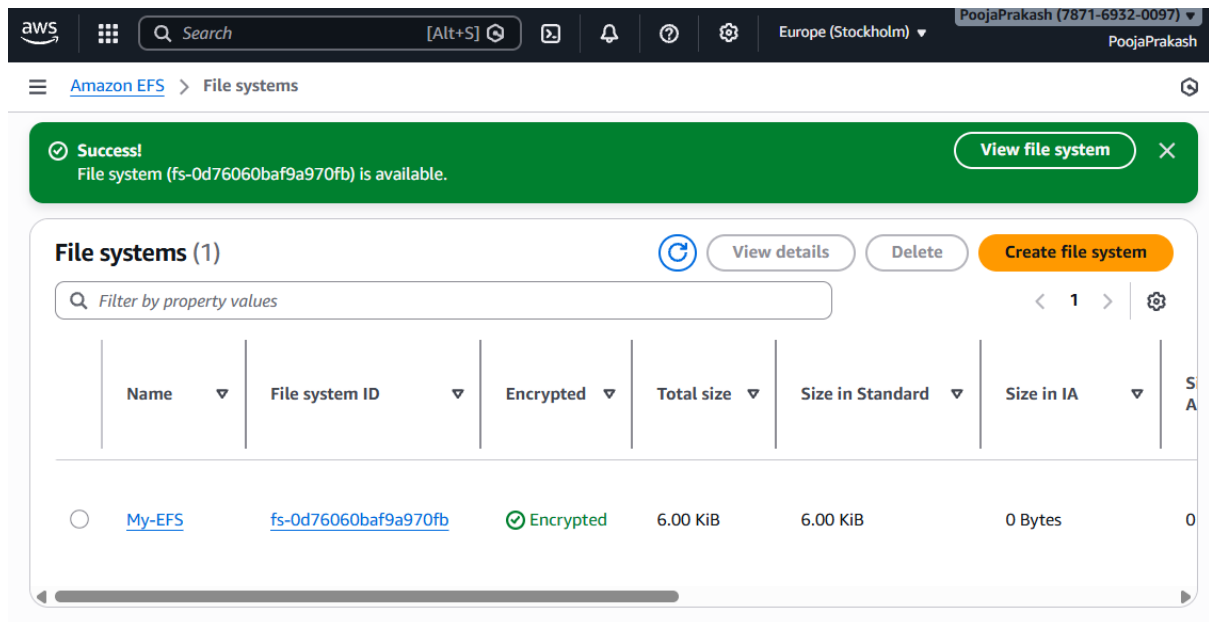
You must delete all existing mount targets in order to change the VPC of your file system.

Mount targets

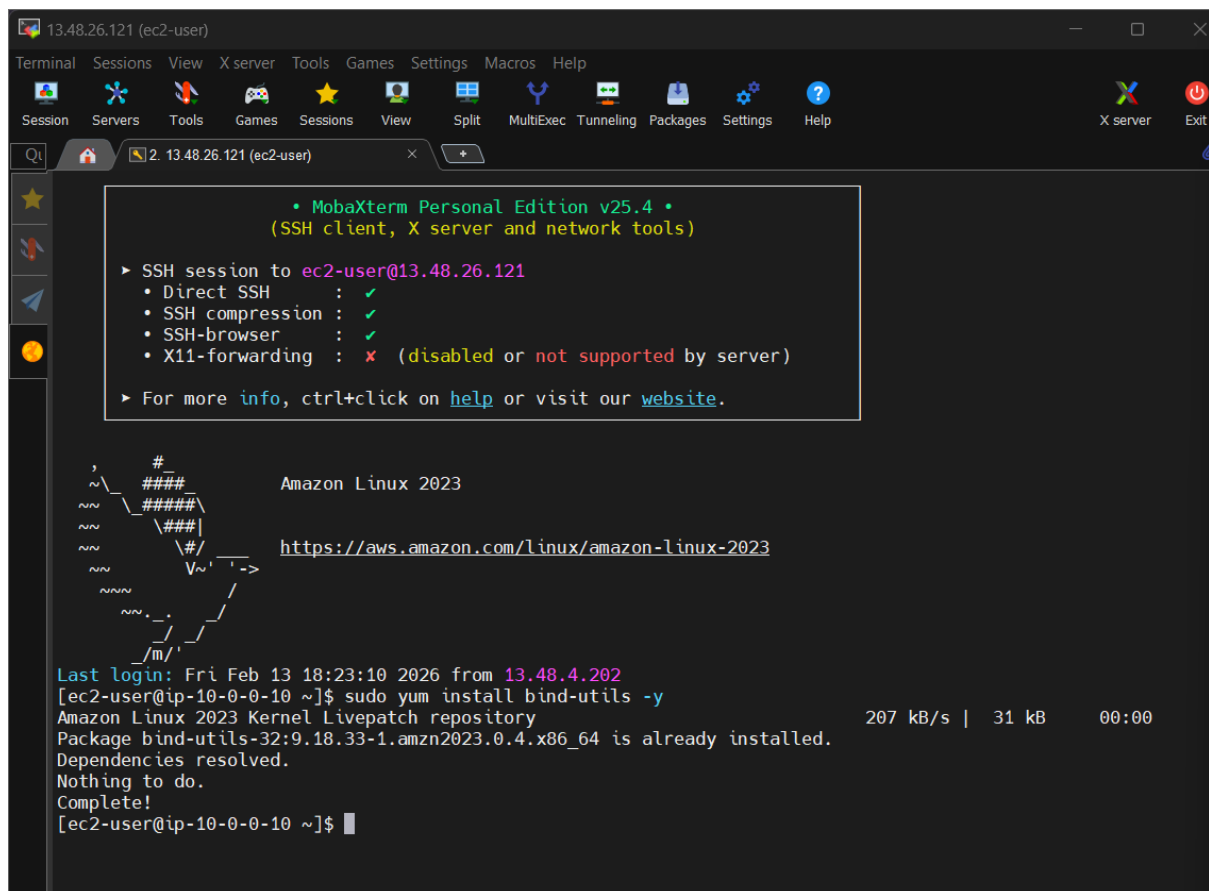
A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#) [↗](#)

Availability zone	Subnet ID	IP address type	IPv4 address	IPv6 address	Security groups	
eu-north-1a	subnet-090dbcf2	IPv4 only	10.0.0.6	-	Choose securi...	Remove
					sg-0d81f329930294ee1 EFS-SG	
eu-north-1b	subnet-02d241831	IPv4 only	10.0.0.21	-	Choose securi...	Remove
					sg-0d81f329930294ee1 EFS-SG	

[Add mount target](#)



- Connect to Instance A, install the required packages and run `dig +short <EFS-DNS-Name>` (the DNS name is in the EFS console).
- This command returns the private IP address through which EFS communicates.



```
[ec2-user@ip-10-0-0-10 ~]$ dig +short fs-0d76060baf9a970fb.efs.eu-north-1.amazonaws.com
10.0.0.6
[ec2-user@ip-10-0-0-10 ~]$
```

- Install the EFS utilities package on Instance A.

```
[ec2-user@ip-10-0-0-10 ~]$ sudo yum install -y amazon-efs-utils
Last metadata expiration check: 0:26:20 ago on Fri Feb 13 18:25:53 2026.
Dependencies resolved.
=====
Package                Architecture    Version           Repository        Size
=====
Installing:
amazon-efs-utils        x86_64         2.4.1-1.amzn2023 amazonlinux        4.7 M
Installing dependencies:
stunnel                 x86_64         5.58-1.amzn2023.0.2 amazonlinux        156 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 4.9 M
Installed size: 10 M
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm           3.6 MB/s | 156 kB      00:00
(2/2): amazon-efs-utils-2.4.1-1.amzn2023.x86_64.rpm     34 MB/s | 4.7 MB      00:00
-----
Total                                                     27 MB/s | 4.9 MB      00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Installing     : amazon-efs-utils-2.4.1-1.amzn2023.x86_64 2/2
  Running scriptlet: amazon-efs-utils-2.4.1-1.amzn2023.x86_64 2/2
  Verifying      : amazon-efs-utils-2.4.1-1.amzn2023.x86_64 1/2
  Verifying      : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2

Installed:
  amazon-efs-utils-2.4.1-1.amzn2023.x86_64          stunnel-5.58-1.amzn2023.0.2.x86_64

Complete!
```

- Create a directory and mount the EFS using the file system ID.

```
[ec2-user@ip-10-0-0-10 ~]$ sudo yum install -y amazon-efs-utils
Last metadata expiration check: 1:09:53 ago on Fri Feb 13 18:25:53 2026.
Package amazon-efs-utils-2.4.1-1.amzn2023.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-0-10 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-10-0-0-10 ~]$ sudo mount -t efs fs-0d76060baf9a970fb:/ /mnt/efs
[ec2-user@ip-10-0-0-10 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0    4.0M   0% /dev
tmpfs           459M   0    459M   0% /dev/shm
tmpfs           184M  444K   183M   1% /run
/dev/nvme0n1p1  8.0G  1.7G   6.4G  21% /
tmpfs           459M   0    459M   0% /tmp
/dev/nvme0n1p128 10M    1.3M   8.7M  13% /boot/efi
tmpfs           92M    0     92M   0% /run/user/1000
127.0.0.1:/      8.0E   0     8.0E   0% /mnt/efs
[ec2-user@ip-10-0-0-10 ~]$
```

- Navigate to the mounted directory and create some files. These files will be stored in the EFS file system.

```
[ec2-user@ip-10-0-0-10 ~]$ cd /mnt/efs
[ec2-user@ip-10-0-0-10 efs]$ sudo bash -c 'echo "Hello from Instance A" > /mnt/efs/test.txt'
[ec2-user@ip-10-0-0-10 efs]$ ls
test.txt
[ec2-user@ip-10-0-0-10 efs]$
```

- Launch another EC2 instance named Instance B in the same VPC, because EFS and EC2 must be in the same VPC in order to communicate.

EC2 > Instances > i-0fe1cfcebe2633e81

Instance summary for i-0fe1cfcebe2633e81 (Instance B) Info

[Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID i-0fe1cfcebe2633e81	Public IPv4 address 13.61.1.236 open address	Private IPv4 addresses 10.0.0.9
IPv6 address -	Instance state Running	Public DNS ec2-13-61-1-236.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-0-9.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-0-9.eu-north-1.compute.internal	
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address 13.61.1.236 [Public IP]	VPC ID vpc-0d32c10ca1a31558f (VPC-EFS)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM role -	Subnet ID subnet-090dbcfa222b1cb6b (EFS-Subnet-01)	Auto Scaling Group name -
IMDSv2 Required	Instance ARN arn:aws:ec2:eu-north-1:787169320097:instance/i-0fe1cfcebe2633e81	Managed false

- Connect to Instance B, install the EFS utilities package, mount the same EFS file system, and verify that the files created from Instance A are visible.

```

#
~\##### Amazon Linux 2023
~\#####|
~\###|
~\#/
~V~'-'>
~m/'-
[ec2-user@ip-10-0-0-9 ~]$ sudo yum install -y amazon-efs-utils
Amazon Linux 2023 Kernel Livepatch repository                250 kB/s | 31 kB    00:00
Last metadata expiration check: 0:00:01 ago on Fri Feb 13 19:52:46 2026.
Dependencies resolved.

=====
Package                                Architecture          Version               Repository            Size
-----
Installing:
amazon-efs-utils                      x86_64                 2.4.1-1.amzn2023     amazonlinux            4.7 M
Installing dependencies:
stunnel                               x86_64                 5.58-1.amzn2023.0.2   amazonlinux            156 k
Transaction Summary
-----
Install 2 Packages

Total download size: 4.9 M
Installed size: 10 M
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm              3.6 MB/s | 156 kB    00:00
(2/2): amazon-efs-utils-2.4.1-1.amzn2023.x86_64.rpm        33 MB/s | 4.7 MB     00:00

[ec2-user@ip-10-0-0-9 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-10-0-0-9 ~]$ sudo mount -t efs fs-0d76060baf9a970fb:/ /mnt/efs
[ec2-user@ip-10-0-0-9 ~]$ cd /mnt/efs
[ec2-user@ip-10-0-0-9 efs]$ ls
test.txt
[ec2-user@ip-10-0-0-9 efs]$ cat test.txt
Hello from Instance A
[ec2-user@ip-10-0-0-9 efs]$

```